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1. A hospital's Medical Superintendent wants to study patient wait time vs staff count across 5 departments. Construct a full additive/multiplicative decomposition for this data, then analyze whether an additive or multiplicative model fits better and justify using the data's variance pattern, and recommend one action for the Medical Superintendent.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

data = [95, 90, 88, 100, 110, 105, 98, 92, 85, 80, 75, 70]

series = pd.Series(data)

# Additive
result = seasonal_decompose(series, model='additive', period=4)
result.plot()
plt.suptitle("Additive Decomposition")
plt.show()

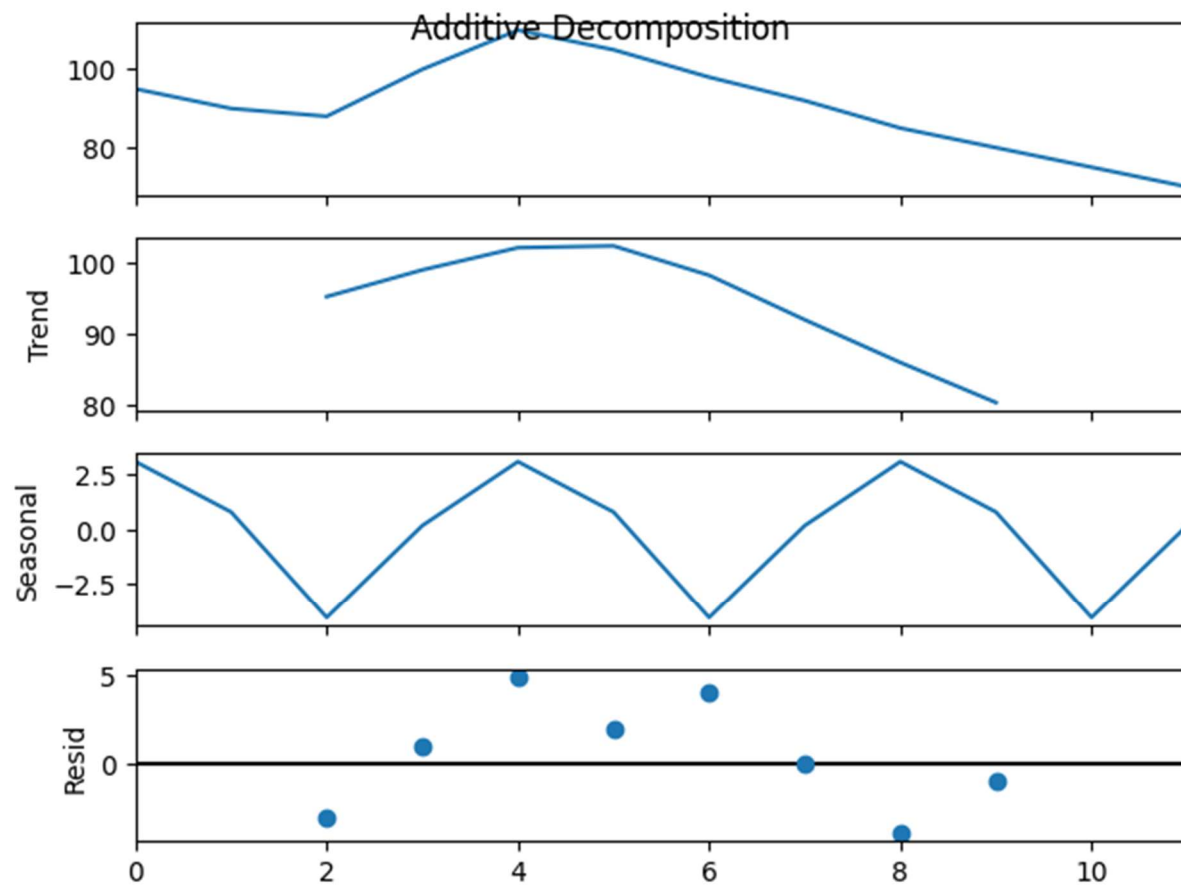
# Multiplicative
result = seasonal_decompose(series, model='multiplicative', period=4)
```

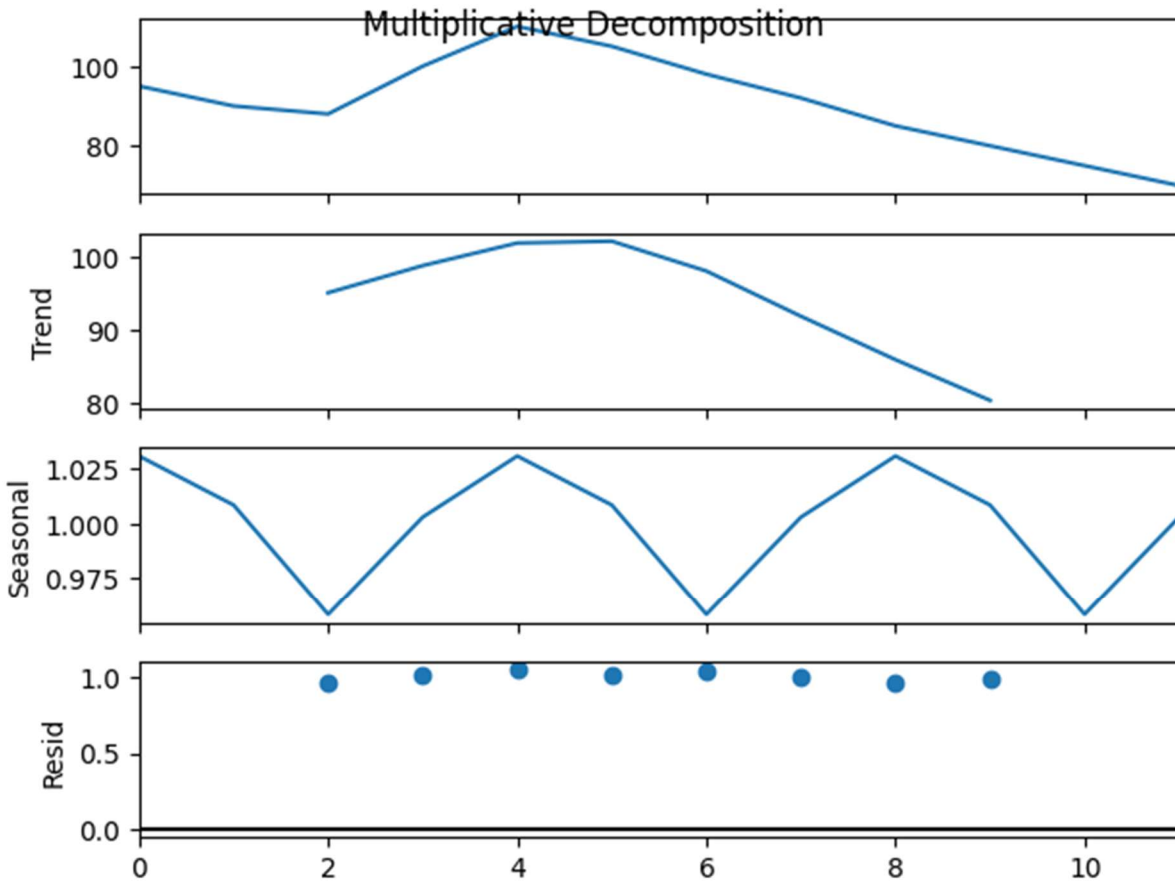
```
result.plot()
```

```
plt.suptitle("Multiplicative Decomposition")
```

```
plt.show()
```

Output:





2. A retail chain's Sales Director wants to study weekly sales (₹ lakhs) vs footfall across 6 store branches. Construct a cartogram resized by a metric, vs the true-area map for this data, then analyze which regions are most visually distorted and what insight the cartogram reveals that the true map hides, and recommend one action for the Sales Director.

Code:

```
import matplotlib.pyplot as plt
```

```
branches = ["Chennai", "Bengaluru", "Hyderabad", "Mumbai", "Pune", "Kochi"]
```

```
sales = [42, 65, 58, 95, 35, 28]
```

```
plt.figure(figsize=(8, 5))
```

```
plt.scatter(branches, sales, s=[x*20 for x in sales])
```

```
for i in range(6):
```

```
    plt.text(branches[i], sales[i], branches[i])
```

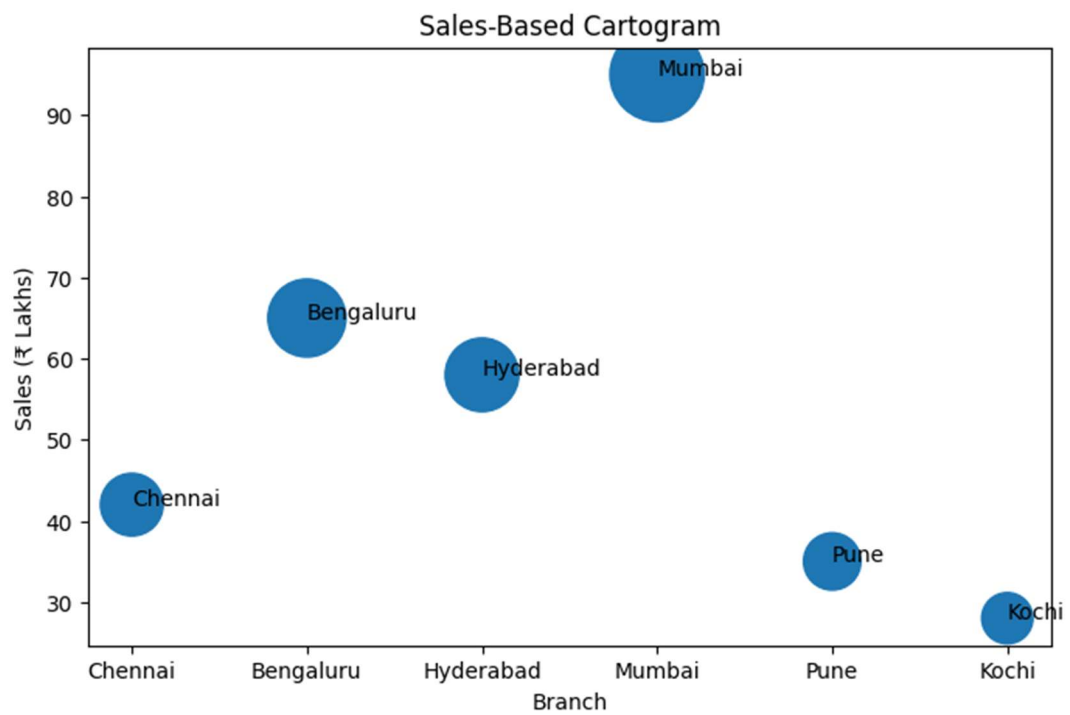
```
plt.title("Sales-Based Cartogram")
```

```
plt.xlabel("Branch")
```

```
plt.ylabel("Sales (₹ Lakhs)")
```

```
plt.show()
```

output:



3. A hospital's Medical Superintendent wants to study patient wait time vs staff count across 5 departments. Construct a 2D projection using PCA and one of t-SNE/UMAP for this data, then analyze how well each technique separates natural groups and explain the variance/neighborhood trade-off between them, and recommend one action for the Medical Superintendent.

Code:

```
import matplotlib.pyplot as plt  
from sklearn.preprocessing import StandardScaler  
from sklearn.decomposition import PCA  
from sklearn.manifold import TSNE
```

```
X = [[12,95],[18,70],[10,120],[15,80],[20,55]]
```

```
X = StandardScaler().fit_transform(X)
```

```
# PCA
```

```
pca = PCA(n_components=2)
```

```
P = pca.fit_transform(X)
```

```
plt.scatter(P[:,0], P[:,1])
```

```
plt.title("PCA 2D Projection")
```

```
plt.show()
```

```
# t-SNE
```

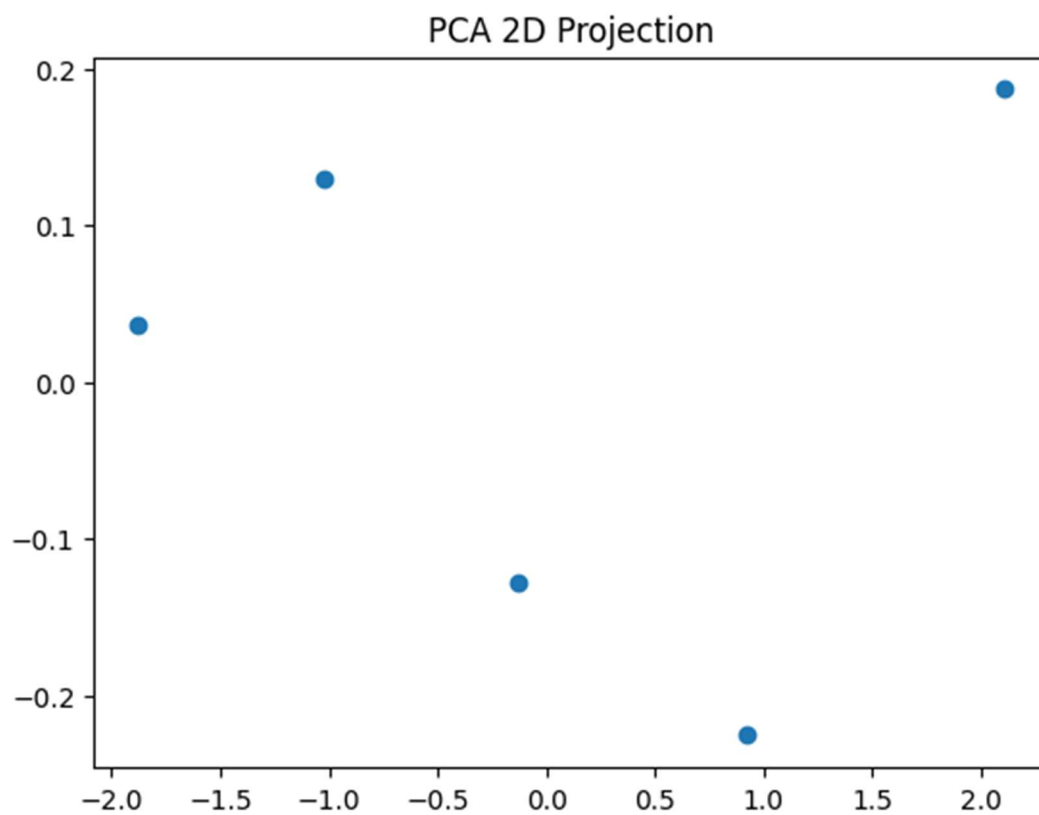
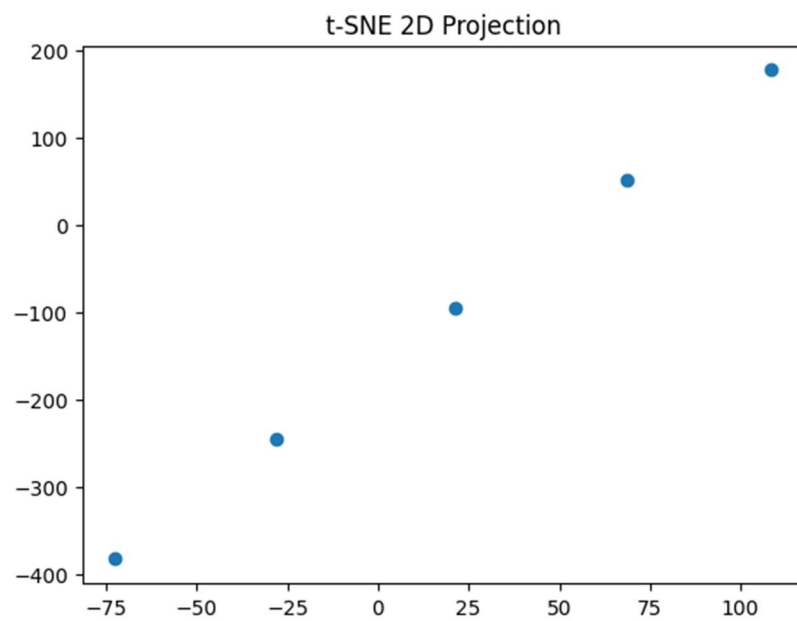
```
T = TSNE(n_components=2, perplexity=2, random_state=42).fit_transform(X)
```

```
plt.scatter(T[:,0], T[:,1])
```

```
plt.title("t-SNE 2D Projection")
```

```
plt.show()
```

output:



4. A hospital's Medical Superintendent wants to study patient wait time vs staff count across 5 departments. Construct a multi-line time series / dose-response chart for this data, then analyze how the series diverge over time or how the response changes with dose, identifying a threshold effect, and recommend one action for the Medical Superintendent.

Code:

```
import matplotlib.pyplot as plt
```

```
staff = [10,12,15,18,20]
```

```
wait = [120,95,80,70,55]
```

```
plt.plot(staff, wait, marker='o')
```

```
plt.axvline(15, linestyle='--')
```

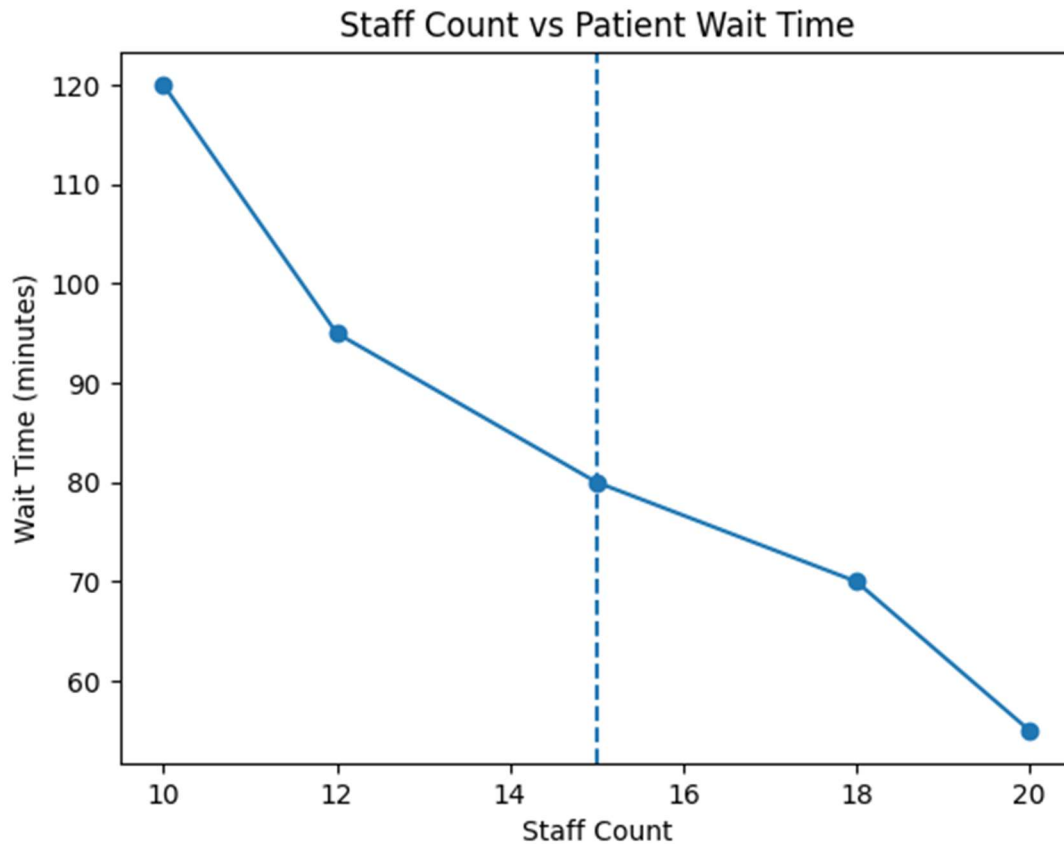
```
plt.title("Staff Count vs Patient Wait Time")
```

```
plt.xlabel("Staff Count")
```

```
plt.ylabel("Wait Time (minutes)")
```

```
plt.show()
```

output:



5. A retail chain's Sales Director wants to study weekly sales (₹ lakhs) vs footfall across 6 store branches. Construct a historical series with a short-horizon forecast and confidence band for this data, then analyze the forecast's uncertainty growth and one risk of over-trusting it, and recommend one action for the Sales Director.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

weeks = np.arange(1,11)
sales = [315,322,330,337,345,351,360,368,375,383]

# Trend line
```

```
coef = np.polyfit(weeks, sales, 1)
```

```
future = np.arange(1,14)
```

```
forecast = np.polyval(coef, future)
```

```
plt.plot(weeks, sales, marker='o', label="Historical")
```

```
plt.plot(future, forecast, '--', label="Forecast")
```

```
plt.axvline(10, linestyle=':')
```

```
plt.title("Weekly Sales Forecast")
```

```
plt.xlabel("Week")
```

```
plt.ylabel("Sales (₹ Lakhs)")
```

```
plt.legend()
```

```
plt.show()
```

output:

